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Date: 2/5/25

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MyDesign Portfolio

Element B Initial Template

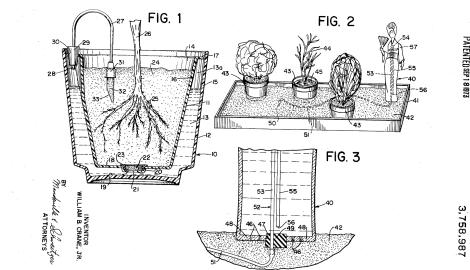
Element B: Documentation and Analysis of Prior Design Attempts

Automatic Plant Watering System via Soil Moisture Sensing:

Source: [https://www.researchgate.net/publication/319130612 Automatic Plant Watering System via Soil Moisture Sensing by means of Suitable Electronics and its Applications for Anthropological and Medical Purposes](https://www.researchgate.net/publication/319130612_Automatic_Plant_Watering_System_via_Soil_Moisture_Sensing_by_means_of_Suitable_Electronics_and_its_Applications_for_Anthropological_and_Medical_Purposes)

In a previous design, an automatic plant watering device was developed to assist in maintaining plant health, particularly for individuals who may forget to water them regularly. This device is equipped with a sensor that detects moisture levels in the soil, allowing it to determine when the plant requires watering. Once the moisture level drops below a certain threshold, the system automatically dispenses water, ensuring that the plant receives the necessary hydration. By automating the watering process, this device helps keep plants healthy and thriving, even in the absence of regular human intervention.

Some early moisture sensors didn't always give accurate readings of how wet the soil really was. This meant that the system could overwater or underwater the plants if the sensor didn't work right.



Same cite as the top source

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Drip Irrigation Systems

Source: https://www.mdpi.com/2073-4441/15/9/1733?utm_source=chatgpt.com

Drip Irrigation Systems: These systems use small tubes to slowly bring water right to the base of each plant. This helps save water by preventing waste from evaporation or runoff. Instead of spraying water everywhere, drip irrigation gives plants just the right amount they need.

One example is the LetPot Automatic Watering System 2.0, which lets you set up watering schedules and control everything through an app. It even tracks water levels, so plants don't get too much or too little water. Drip irrigation is great for indoor plants, gardens, and greenhouses because it keeps plants healthy while making watering easier.

One issue that was often found with earlier drip irrigation systems is that they require precise setup and maintenance. If the system is not installed correctly or if the tubes get clogged, the plants will not receive the correct amount of water.

Drip irrigation systems, while efficient, don't always allow for easy adjustments based on plant needs. For example, plants of different sizes or species may need different amounts of water, and sometimes Drip irrigation can't provide that .

Drip irrigation systems can achieve over 90% efficiency in water usage



<https://extension.colostate.edu/topic-areas/yard-garden/drip-irrigation-home-gardens-4-702/?utm>

<https://letpot.com/>

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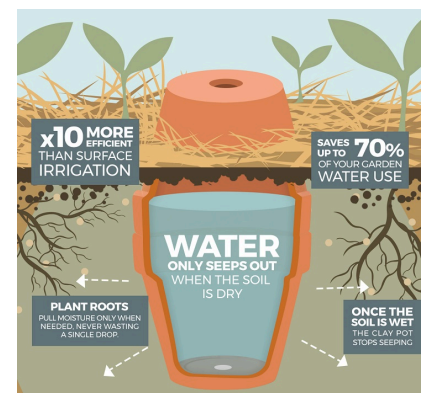
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Olla irrigation Pots

Source <https://extension.arizona.edu/sites/extension.arizona.edu/files/pubs/az1911-2021.pdf>

in-text citations

Olla irrigation uses unglazed clay pots that are buried in the ground to water plants. These pots are special because they let water slowly leak through them into the soil around the plant roots. This method saves a lot of water compared to regular watering. Olla irrigation works based on a simple idea. When the clay pot is filled with water, the soil around the pot pulls water out of it to make the soil moist. As long as the pot is kept full, the plants will always have enough water. This system is easy to use and inexpensive, and it works well for plants like vegetables and flowers.



<https://www.quickcrop.ie>

However, it might not be great for bigger plants like trees /because their roots can break the pots. Olla irrigation also needs space in the garden, which can be a problem if you have a small garden. The pots need to be refilled by hand, which can take a lot of time, especially in hot weather. If you forget to refill it, the plants might dry out.

Olla pots can reduce water usage by up to 70% compared to traditional watering methods.- <https://www.bhg.com/what-is-olla-pot-8669101?utm>

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A Shaduf

Source: <https://www.britannica.com/technology/shaduf>



The Shaduf (circa 1500 BC) is used in Egypt and Mesopotamia, this lever - and bucket system allowed for lifting water from rivers and canals to irrigate crops. Archaeological evidence suggests even earlier variations. The Shaduf has done a terrific job with helping the plants with all the Water Delivery, Enhanced Crop Yields, Soil Moisture Regulation, Erosion Control, Minimal Environment Impact, and the plants are able to adapt to local conditions.

The Shaduf represents a significant advancement in ancient irrigation technology, directly impacting plant health and agricultural productivity. Its ability to efficiently deliver water to crops enhanced food security and supported growth of civilization. By looking at the Shaduf through the lens of plants and water, we gain insight into the interconnectedness of technology, agriculture, and society.

Some of the Downfalls for Shaduf are the time constraints, Continuous Manual Operation, Low Water Delivery rate, and Proximity of water bodies. These limitations spurred the development of more advanced irrigation technologies over time.

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Prior Design Attempt #5 - <Source (Caso, A. (1982). Plant waterers (4,336,666). Patent and Trademark Office.) #5>

"The reservoir has three metering nozzles with perforated holes, which are pressed into the soil to reach deeper parts of the soil. " This could work because the nozzles could reach any part of

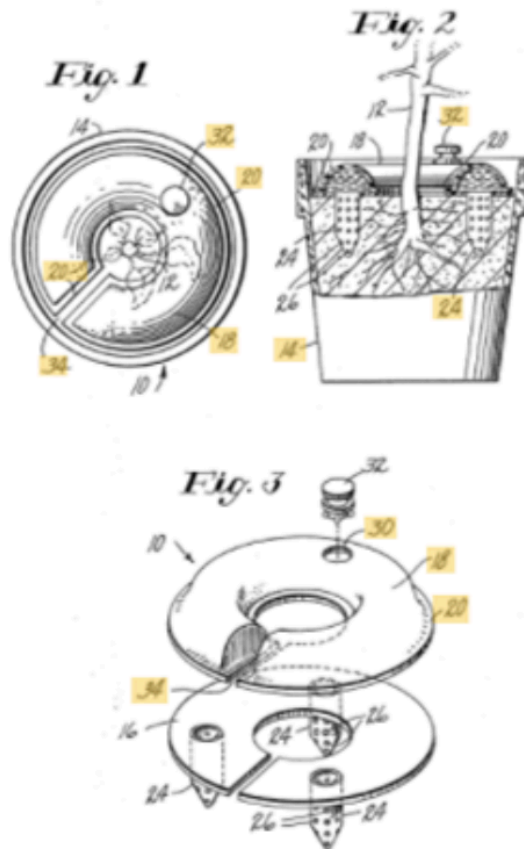
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the soil through one of the nozzles and a con is there would be no way to tell when the reservoir

U.S. Patent Jun. 29, 1982 4,336,666



is empty.

Details

It has 3 nozzles that are each used to reach a different part of the soil. That's very helpful because sometimes the soil could absorb the water if you pour it from the top.

Analysis

This design was unsuccessful at having a very aesthetic look for the plant, there would be no real way to tell when the reservoir was empty and if you leave the reservoir unattended it may turn into a habitat for mosquitoes and pests. To solve the aesthetic look you could try to hide the nozzles, and to tell how much water is left, make a sign or something that will show how much

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percent is left and for the pest and mosquitoes you could use a trap for them to get rid of them and keep them away from your plant .